Run report: run-20261011T2258Z-glvgls-lattice

What ran. `cargo run --release --example isogeny_loop_glv_gls_lattice -- --bits 127 --samples 10000 --seed 20261011 --max-tries 2000000 --discs -3,-4,-7,-8,-11,-23,-71,-431,-619,-3019 --json research/isogeny_loop_glv_20261011/glvgls_lattice.json`. Base commit db0393b4 plus the example source with SHA-256 `e789d26c…4c01` (`../../revision.txt`); rustc 1.97.0; Linux 6.18 x86_64 cloud container, not isolated. Start 2026-10-11T22:58Z (`../../glvgls_run_start_utc.txt`); wall time 7.6 s (`time`, recorded in `../../glvgls_lattice.log`). Seed 20261011 (per-discriminant streams derived from it).

Arms. Per discriminant: certified cheapest loop (exhaustive to the norm bound); GLS-2 lattice; GLV-GLS-4 lattice for both ring embeddings (s, $-s$). 10,000 seeded scalars per lattice.

Runs requested / completed. 10 discriminants requested; 9 produced an instance; $D = -4$ produced none within $2 \cdot 10^6$ draws (recorded; the cause is derived in the study README §3). **Earlier runs:** a 22:51Z run of the same source before formatting and clippy-only edits produced identical rows (checked field by field). **Failed earlier attempt, kept:** the first full run, without the draw cap, was stopped by hand after 13.5 CPU-minutes stuck in the $D = -4$ search; it wrote no records. Checks asserted per instance: $\mu^2 = -1$, $\mu^2 - t\mu + p = 0$, $s^2 = D$, $\lambda^2 - \mathrm{Tr}(\alpha)\lambda + N(\alpha) = 0 \pmod r$; every LLL vector in its lattice; every decomposition congruent to its scalar. All passed (exit 0).

Recorded values (unit: bits, max / mean over 10,000 scalars of $\max_i \mathrm{bits}(|k_i|)$): GLS-2 124–126 / 122.8–125.5; GLV-GLS-4 62–63 / 61.2–62.6, against $\log_2 r/4$ = 62.0–63.5. Cheapest loop costs (M, projective): 1, 1, 15, 15, 22.5, 37.5, 60, 97.5, 127.5, 210. Model ratio GLS-2 / GLV-GLS-4 (derived operation count, not measured): 1.158–1.357. Full table: `../../glvgls_lattice.md`; record: `../../glvgls_lattice.json`.

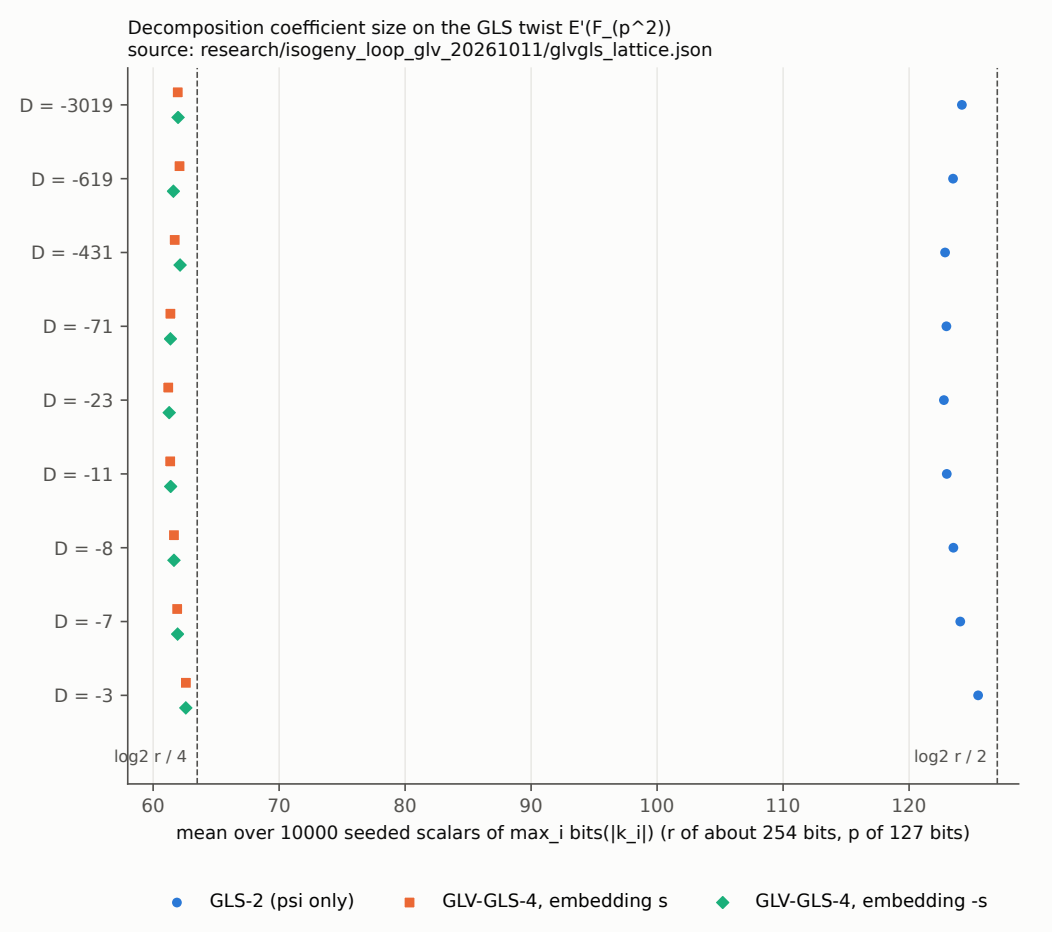


Figure 1: Recorded mean coefficient sizes per discriminant (run run-20261011T2258Z-glvgls-lattice).

Observations only. Interpretation is in `../../README.md` §3 and `../../REPORT.pdf`.